in both forms, also exhibit alternation of generations in their life cycle (**Metagenesis**). The polyp represents the asexual generation and medusa represents the sexual generation. Polyps produce medusa asexually and medusa forms polyps sexually. Development is indirect and includes a free swimming ciliated **planula larva**.

Examples: *Physalia* (Portugese man of war), *Adamsia* (Sea anemone), Pennatula (Sea pen), *Meandrina* (Brain coral) (Figure 2.10).

Compare the advantages and disadvantages of direct and indirect development.

2.3.3 Phylum: Ctenophora

(G. *Ktenos* -comb; *phoros* -bearing)

Ctenophora are exclusively marine, biradially symmetrical, diploblastic animals with tissue level of organisation. Though they are diploblastic, their mesoglea is different from that of cnidaria. It contains amoebocytes and smooth muscle cells. They have eight external rows of ciliated comb plates (comb jellies) which help in locomotion, hence commonly called **comb jellies** or **sea walnuts**. **Bioluminescence** (the ability of a living organism to emit light) is well marked in ctenophores. They lack nematocysts but possess special cells called **lasso cells** or **colloblasts** which help in food capture. Digestion is both extracellular and intracellular. Sexes are not separate (monoecious). They reproduce only by sexual means. Fertilization is external and development is indirect and includes a larval stage called **cydippid larva**. e.g., *Pleurobrachia* (Figure 2.11).

Examples : *Pleurobrachia and Ctenoplana*.

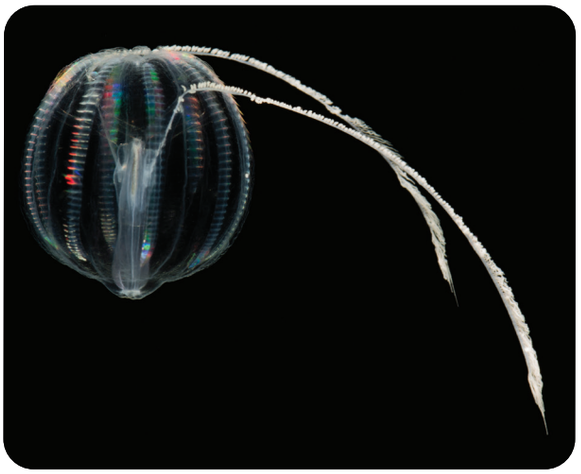


Figure 2.11 Example of Ctenophora- *Pleurobrachia*

2.3.4 Phylum: Platyhelminthes (Flatworms)

(G. *Platy* -broad or flat; *helmin*-worm)

They have a dorsoventrally flattened body and hence called flatworms. These animals are bilaterally symmetrical, triploblastic, acoelomate with organ system level of organisation. They show moderate cephalization and unidirectional movement. They are, mostly endoparasites of animals including human beings. Hooks and suckers are present in the parasitic forms and serve as organs of attachment. Their body is not segmented, but some exhibit pseudosegmentation. Some of the parasitic flatworms absorb nutrients directly from the host through their body surface. However, flatworms like liver fluke have an incomplete digestive system. Specialized excretory cells called **flame cells** help in osmoregulation and excretion. Sexes are not separate (monoecious); fertilisation is internal and development is through larval stages (miracidium, sporocyst, redia, cercaria). **Polyembryony**

XI Std Zoology Chapter-2 EM.indd 21 30/12/2022 10:30:55